SAMPLE SLIDES

Mississippi State · Ph.D. research

High-Speed PMSM Control Drives

A Direct Torque Control drive for a fuel-cell compressor motor on a 200 kW, 450 V inverter, validated on hardware to 60,000 rpm.

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Problem & motivation

Fuel-cell systems use high-speed compressors to supply air, and the motors that drive them must start reliably and hold speed under changing load. At standstill there is no reliable flux or position information, so a high-speed DTC drive needs a safe way to start before closed-loop control can take over.

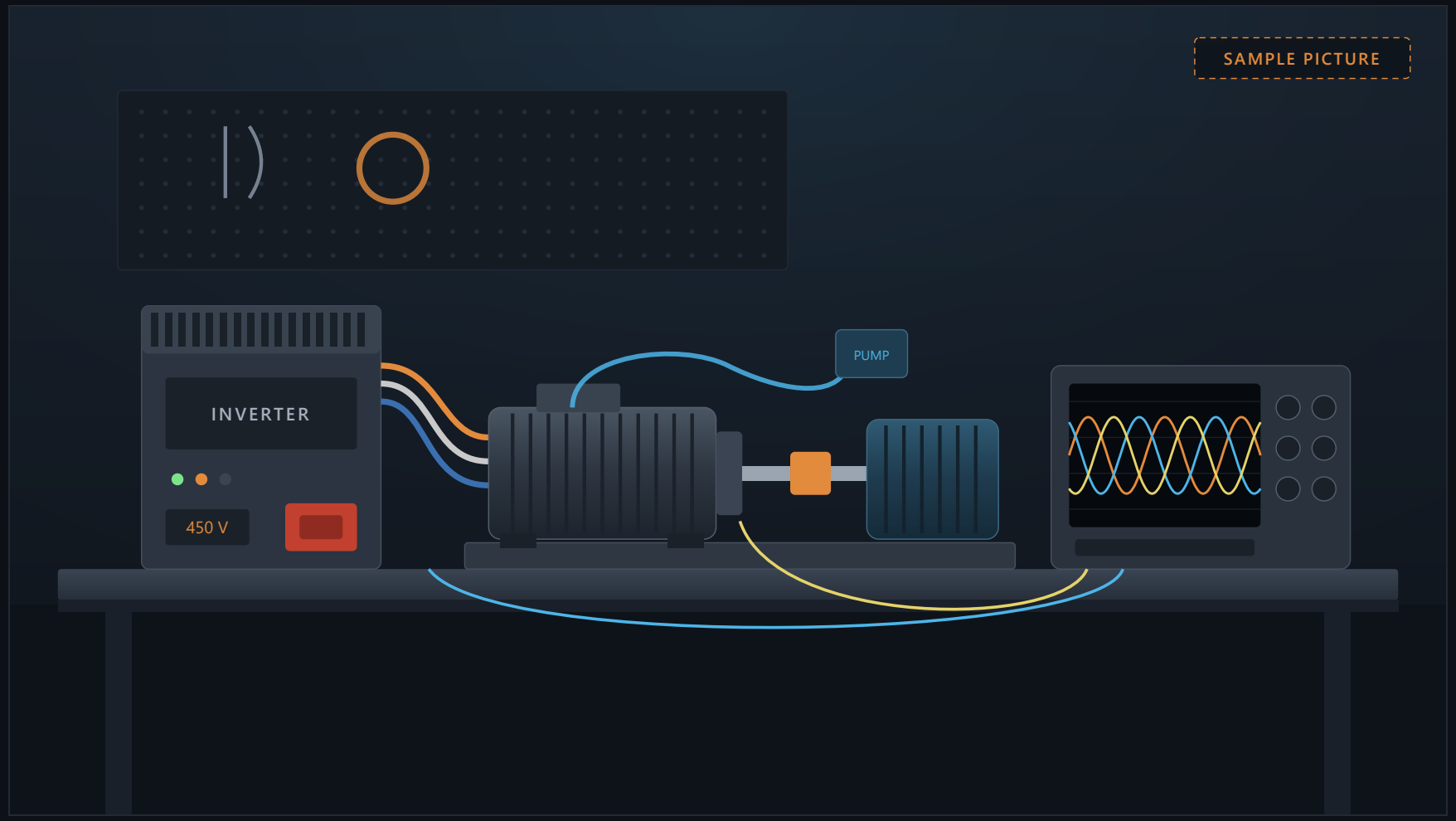
Because the drive is high-power and high-speed, heat and protection matter as much as the control law.

Approach

- Developed DTC with a seamless open-loop to closed-loop transition strategy
- Designed and tuned PI controllers for stable closed-loop operation under varying load
- Integrated the high-power inverter with the compressor motor
- Engineered active cooling to keep the system stable and prevent overheating at high speed

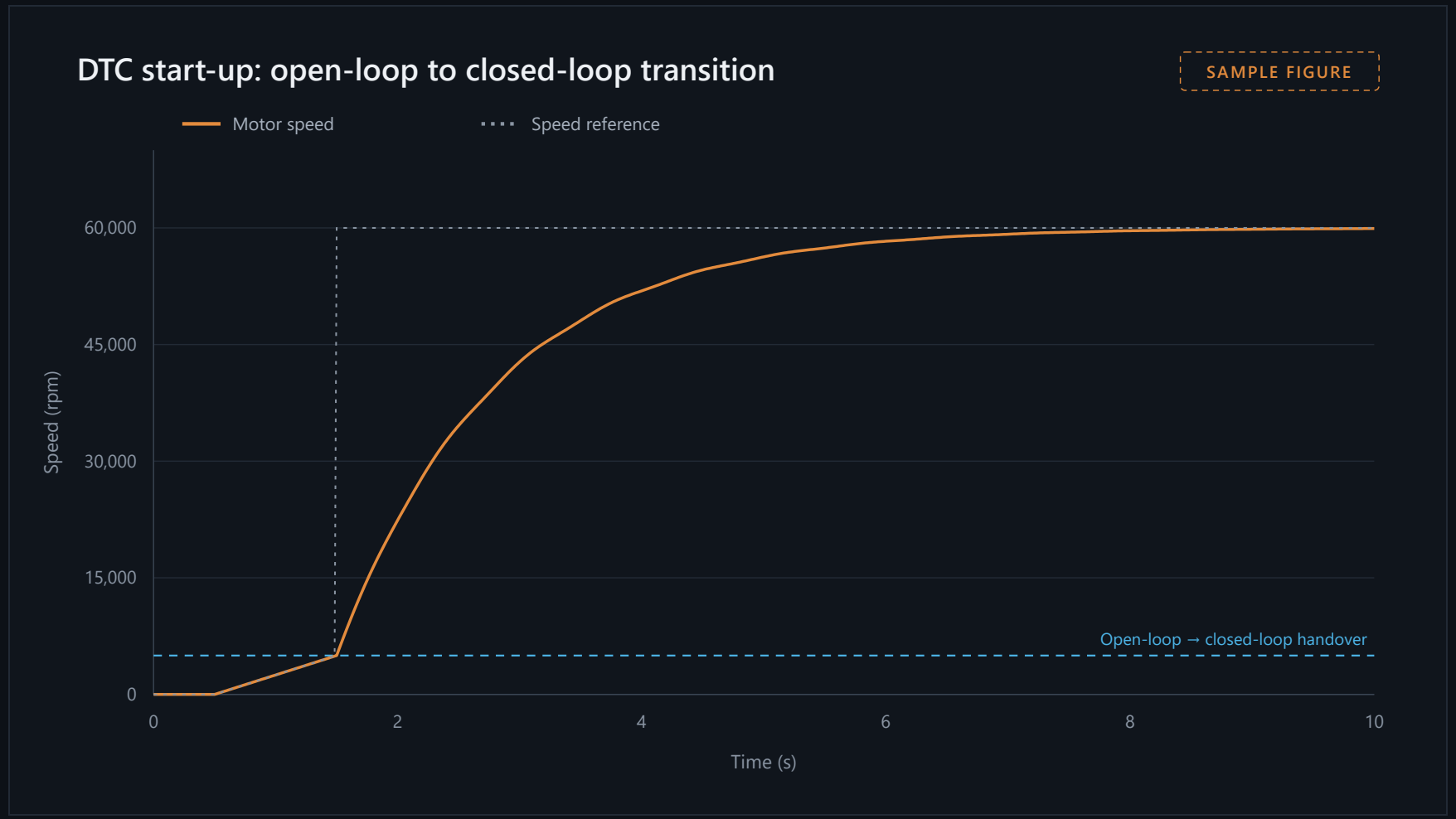
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Motor test bench: inverter, high-speed PMSM coupled to a load, active cooling, and oscilloscope.



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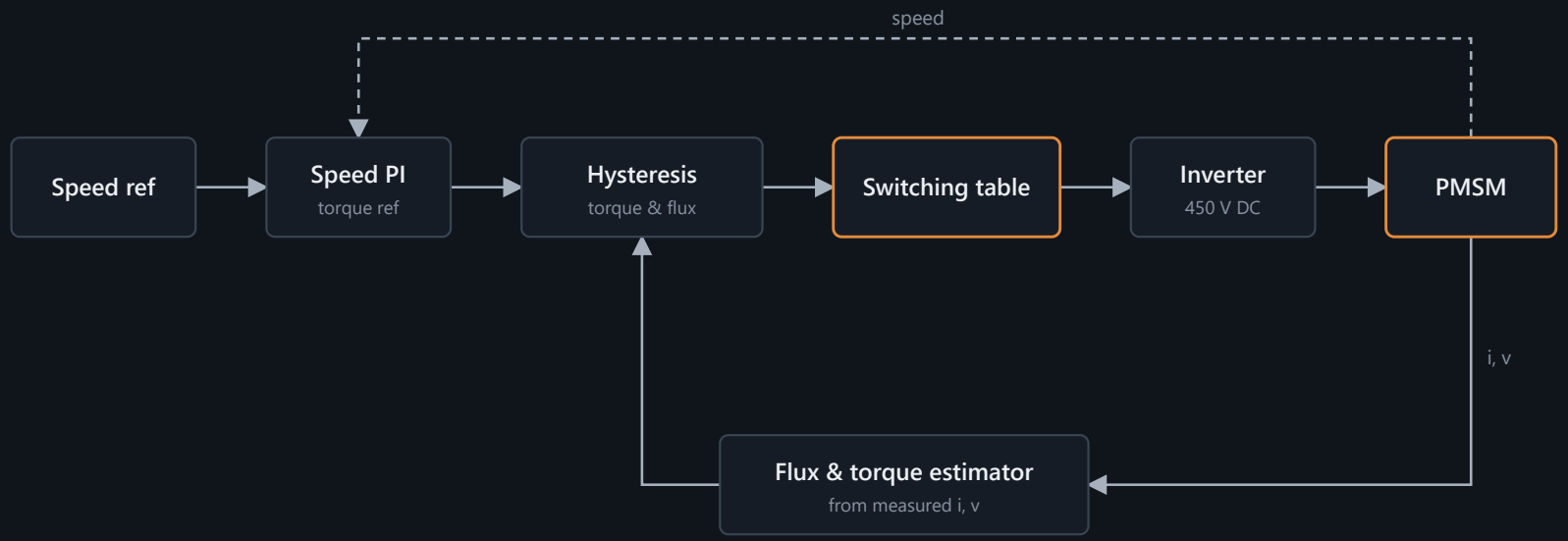
Start-up with the open-loop to closed-loop transition, up to 60,000 rpm.



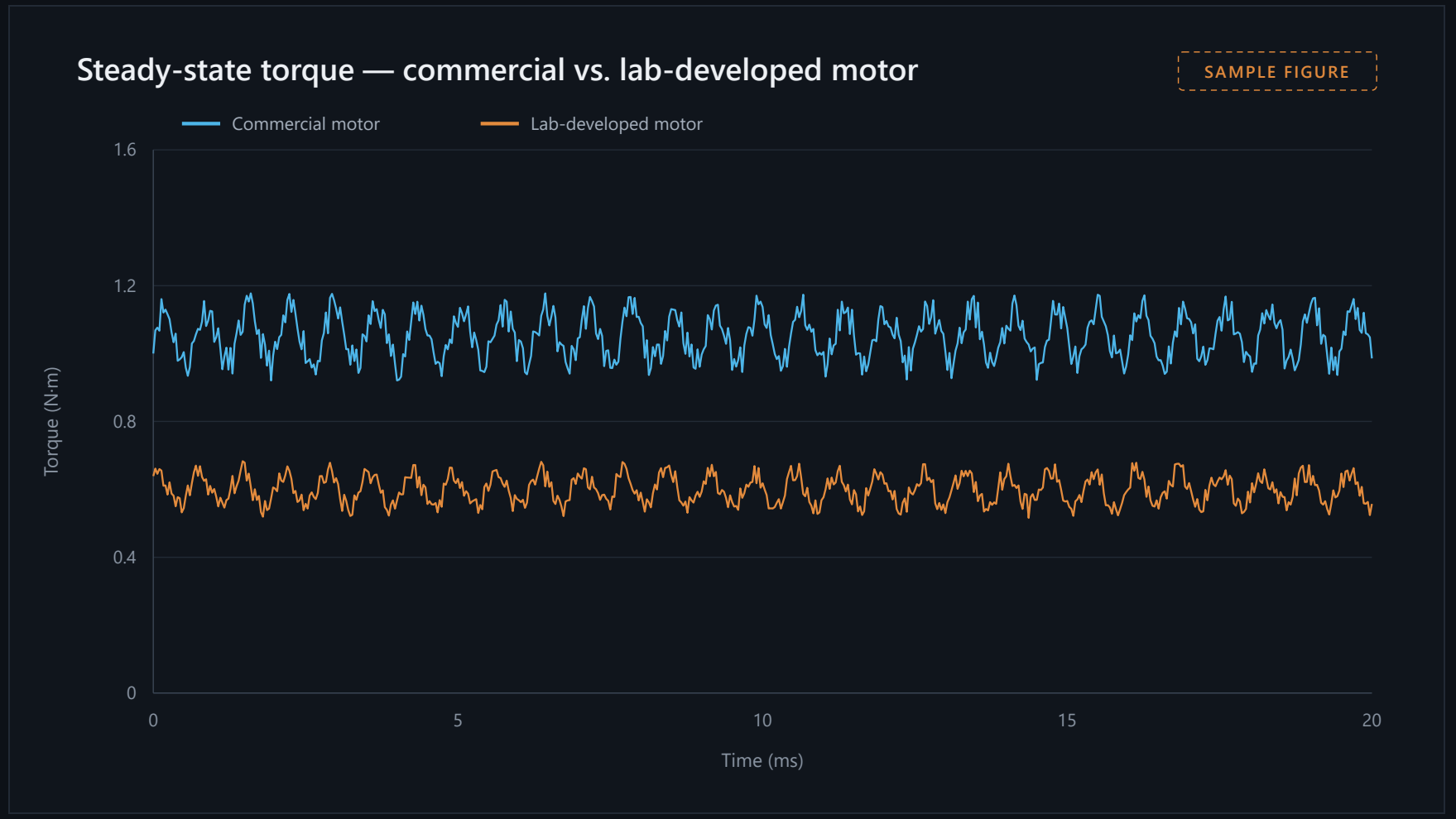
DTC structure: hysteresis torque and flux control with a switching table.

Direct Torque Control structure

SAMPLE FIGURE



Steady-state torque of the commercial and lab-developed motors.



Results

60,000 rpm	200 kW	450 V	DTC
Validated on hardware	XM3 inverter	DC bus	Control method

- Validated on hardware up to 60,000 rpm
- Stable high-speed performance under varying load conditions
- Published comparison of DTC on commercial and lab-developed PMSMs at IEEE APEC 2025

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Thank you

Questions and discussion

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